



Roll No.

Total No. of Questions : 31]

[Total No. of Printed Pages : 8

12thARF(SZ)JKUT2024-25

307-Z

MATHEMATICS

Time : 3 Hours]

[Maximum Marks : 80

SECTION-A

**(OBJECTIVE TYPE QUESTIONS/
MULTIPLE CHOICE QUESTIONS)**

1 each

1. If

$$\begin{vmatrix} x & -2 \\ 6 & 4 \end{vmatrix} = 4$$

then x is equal to :

(A) -2

(B) 2

(C) 6

(D) 4

2. A square matrix A is singular if $|A| = \dots\dots\dots$

12thARF(SZ)JKUT2024-25-307-Z

B-7-Z

Turn Over

3. Derivative of $e^{\log \sin x}$ w.r.t. x is :
- (A) $\cos x$
- (B) $-\cos x$
- (C) $\sin x$
- (D) $-\sin x$
4. Second derivative of x^{20} w.r.t. x is :
- (A) $380 x^{18}$
- (B) $190 x^{18}$
- (C) $20 x^{18}$
- (D) $19 x^{18}$
5. The function $f(x) = 5x - 3$ is continuous at $x = 0$. (True/False)
6. Maximum value of $\sin 2x + 5$ is :
- (A) 5
- (B) 4
- (C) 6
- (D) 7

7. $\int \sec x \, dx$ is equal to :
- (A) $\log (\sec x + \tan x) + C$
- (B) $\log (\sec x - \tan x) + C$
- (C) $\log (\cot x + \operatorname{cosec} x) + C$
- (D) $\log (\cot x + \operatorname{cosec} x) + C$
8. If two lines are intersecting, then shortest distance between them =
9. Projection of $i + \hat{j}$ on $\hat{i} - \hat{j}$ is zero. (True/False)
10. Define feasible region.

SECTION-B

(VERY SHORT ANSWER TYPE QUESTIONS) 2 each

11. Check whether the relation R defined in the set $\{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) \mid b = a + 1\}$ is reflexive, symmetric or transitive.
12. Prove that :

$$\sin^{-1} (2x \sqrt{1-x^2}) = 2 \sin^{-1} x$$

$$-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$$

13. Find the rate of change of the area of a circle with respect to its radius ' r ' when $r = 3$ cm.

14. Find $|\vec{a}|$ and $|\vec{b}|$, if

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$$

and

$$|\vec{a}| = 8|\vec{b}|$$

15. If

$$\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$$

and

$$\vec{b} = -\hat{i} + \hat{j} - \hat{k}$$

find the unit vector in the direction of $\vec{a} + \vec{b}$.

16. Find :

$$\int \frac{\cos x}{\sqrt{1 + \sin x}} dx$$

17. Evaluate :

$$\int_0^{\pi/4} \tan x dx$$

18. A coin is tossed three times. If E is the event 'head on third toss' and F is the event 'heads on first two tosses'. Find $P(E|F)$.
19. Let E and F be two events with $P(E) = \frac{3}{5}$, $P(F) = \frac{3}{10}$ and $P(E \cap F) = \frac{1}{5}$. Are E and F independent ?
20. Find the product :

$$\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

SECTION-C

(SHORT ANSWER TYPE QUESTIONS)

4 each

21. Find the relationship between a and b so that the function f defined by <https://www.jkbboseonline.com>

$$f(x) = ax + 1 \text{ if } x \leq 3$$

$$= bx + 3 \text{ if } x > 3$$

is continuous at $x = 3$.

22. Evaluate :

$$\int_0^1 x(1-x)^n dx$$

23. Differentiate $(\log x)^x + x^{\log x}$ w.r.t. x .

24. Find the shortest distance between the lines :

$$\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$$

and

$$\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$$

25. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that :

$$\vec{a} + \vec{b} + \vec{c} = 0$$

find the value of

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$$

26. Solve the linear programming problem graphically :

$$\text{Maximize } z = 3x + 2y$$

Subject to :

$$x + 2y \leq 10,$$

$$3x + y \leq 15,$$

$$x, y \geq 0.$$

17. Let A and B be two sets. Show that :

$$f : A \times B \rightarrow B \times A$$

such that $f(a, b) = (b, a)$ is bijective function.

18. An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and is returned to the urn. Moreover, 2 additional balls of the colour drawn are put in the urn and then a ball is drawn at random. What is the probability that the second ball is red ?

SECTION-D

(LONG ANSWER TYPE QUESTIONS)

6 each

19. For a matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

show $A^3 - 6A^2 + 5A + 11I = 0$.

Or

Solve the system of linear equations, using matrix method :

$$2x + 3y + 3z = 5$$

$$x - 2y + z = -4$$

$$3x - y - 2z = 3$$

30. Evaluate :

$$\int_0^{\pi/2} \sqrt{\sin \phi} \cos^5 \phi d\phi$$

Or

Evaluate :

$$\int \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx$$

31. Find the local maxima and local minima, if any, of the function :

$$f(x) = \sin x - \cos x, 0 < x < 2\pi.$$

Or

Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.